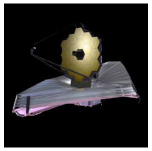

# 12494 - Characterizing a Luminous Red Nova in Real Time

Cycle: 4, Proposal Category: DD

**INVESTIGATORS**



| *Name* | *Institution* |
|---|---|
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**OBSERVATIONS**

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| obs | | | | |
| | 1 | MIRI-LRS | MIRI Low Resolution Spectroscopy | (1) AT2025abao |
| | 3 | MIRI-LRS_epoch2 | MIRI Low Resolution Spectroscopy | (1) AT2025abao |
| | 2 | NIRSPEC | NIRSpec Fixed Slit Spectroscopy | (1) AT2025abao |
| | 4 | NIRSPEC_epoch2 | NIRSpec Fixed Slit Spectroscopy | (1) AT2025abao |

**ABSTRACT**

A binary merger is a key missing phase in our understanding that characterizes the stellar evolution, directly linked to various unresolved topics such as the formation of compact-object binaries. Luminous Red Novae (LRN) are believed to announce mergers of binary stars, which are usually identified by outburst in the optical wavelengths. AT2025abao is a new LRN discovered in M31 - the first LRN discovered in such a proximity after the launch of JWST, for which real-time monitoring of Infra-Red (IR) natures in the entire evolution will be possible. Its importance is further highlighted by the detection of a pre-burst progenitor system. AT2025abao thus represents a unique and ideal target for JWST follow-up observations among LRNe. With this DD proposal, we will perform JWST follow-up observations of AT2025abao in two epochs; in the current visibility window (until Jan 2026; 1-3 months since the LRN optical outburst) and in the next visibility window (Jul 2026 - Sep 2026; 9-11 months). Tracking the IR evolution in the first year of the LRN burst represents a previously unexplored regime. This will be done with the total observing time of ~5.7 hrs for

the two epochs, including overhead. We aim at characterizing the nature of dusty circumstellar environment around an LRN in time sequence, with which we will investigate the expected change in the origins of the dust emissions as an echo by existing dust shells ejected in pre-outburst mass loss to newly formed dusts within the post-outburst ejecta; this will thus provide key insight on when and how the materials are ejected in a binary merger, and how the merger outcome evolves into the post-merger "water fountain" system.

**OBSERVING DESCRIPTION**

We propose observations of AT2025abao with NIRSpec and MIRI LRS, in two epochs; the first observation in the current visibility window until 23 Jan 2026 (1-3 months after the optical outburst), and the second one in the next visibility window (14 Jul 2026 -- 16 Sep 2026; 9-11 months).

The set up is as follows:

MIRI LRS slit spectroscopy with the wavelength coverage of 5 to 14 micron, so that all the emission features of interest in this proposal are covered. The exposure is performed with a combination of 10 groups / integration, 10 integrations / exposure, and 2 total dithers, using the FASTR1 readout pattern, resulting in an exposure time of ~600 s. It will take ~1.2 hrs including the overheads, for a single epoch.

NIR spectroscopy uses high-resolution gratings; G140H, G235H, and G395H, covering 1-5 micron, aiming at resolving various metal and molecule lines. For each setting, the exposure is performed with a combination of 10 groups / integration, 5 integrations / exposure, and 2 total dithers, resulting in an exposure time of ~600 s for each. Including the overhead, the NIR spectroscopy will take ~1.64 hrs for a single epoch.

In total, we request 2.84 hrs to complete the proposed observation at a single epoch, and thus the total request is 5.7 hrs including the overheads.

The expected fluxes across the MIRI LRS and NIR spectroscopy wavelength coverage allow sufficient SNRs for our scientific objectives (for details, see TJ in the attached pdf).

Proposal 12494 - Targets - Characterizing a Luminous Red Nova in Real Time

**Fixed Targets**

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | AT2025abao | RA: 00 38 48.6040 (9.7025167d)<br>Dec: +40 46 8.20 (40.76894d)<br>Equinox: J2000 | Epoch of Position: 2016 | |

*Comments:*
*Category=Star*
*Description=[Eruptive variables, K giants, M stars]*
*Extended=NO*

## Observation

**Proposal 12494, Observation 1: MIRI-LRS**

Thu Dec 18 20:00:32 GMT 2025

**Diagnostic Status: Warning**

Observing Template: MIRI Low Resolution Spectroscopy

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | AT2025abao | RA: 00 38 48.6040 (9.7025167d)<br>Dec: +40 46 8.20 (40.76894d)<br>Equinox: J2000 | Epoch of Position: 2016 | |

*Comments:*
*Category=Star*
*Description=[Eruptive variables, K giants, M stars]*
*Extended=NO*

## Acquisition

| # | Target | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 AT2025abao | F560W | FAST | 4 | 1 | 1 | 11.1 | 281633 |

## Template

| Subarray | Obtain Verification Image? |
|---|---|
| FULL | true |

## Dithers

| # | Dither Type | No. Spectral Steps | Spectral Step Offset | No. Spatial Steps | Spatial Step Offset |
|---|---|---|---|---|---|
| 1 | ALONG SLIT NOD | | | | |

## Pointing Verification

| # | PV Readout Pattern | PV Groups/Int | PV Integrations/Exp | PV Total Integrations | PV Exposures/Dith | PV Total Dithers | PV Total Exposure Time | Optional ETC ID | Filter |
|---|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 5 | 1 | 1 | 1 | 1 | 13.875 | | F560W |

Proposal 12494 - Observation 1 - Characterizing a Luminous Red Nova in Real Time

**Spectral Elements**

| # | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Exposures/Dith | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 10 | 10 | 20 | 1 | 2 | 604.959 | 281633 |

**Special Requirements**

Between Dates 22-NOV-2025:00:00:00 and 23-JAN-2026:00:00:00

## Observation

**Proposal 12494, Observation 3: MIRI-LRS_epoch2**

Thu Dec 18 20:00:32 GMT 2025

**Diagnostic Status: Warning**

Observing Template: MIRI Low Resolution Spectroscopy

## Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | AT2025abao | RA: 00 38 48.6040 (9.7025167d)<br>Dec: +40 46 8.20 (40.76894d)<br>Equinox: J2000 | Epoch of Position: 2016 | |

*Comments:*
*Category=Star*
*Description=[Eruptive variables, K giants, M stars]*
*Extended=NO*

## Acquisition

| # | Target | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 AT2025abao | F560W | FAST | 4 | 1 | 1 | 11.1 | 281633 |

## Template

| Subarray | Obtain Verification Image? |
|---|---|
| FULL | true |

## Dithers

| # | Dither Type | No. Spectral Steps | Spectral Step Offset | No. Spatial Steps | Spatial Step Offset |
|---|---|---|---|---|---|
| 1 | ALONG SLIT NOD | | | | |

## Pointing Verification

| # | PV Readout Pattern | PV Groups/Int | PV Integrations/Exp | PV Total Integrations | PV Exposures/Dith | PV Total Dithers | PV Total Exposure Time | Optional ETC ID | Filter |
|---|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 5 | 1 | 1 | 1 | 1 | 13.875 | | F560W |

# Proposal 12494 - Observation 3 - Characterizing a Luminous Red Nova in Real Time

## Spectral Elements

| # | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Exposures/Dith | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 10 | 10 | 20 | 1 | 2 | 604.959 | 281633 |

## Special Requirements

Between Dates 14-JUL-2026:00:00:00 and 16-SEP-2026:00:00:00

## Observation

**Proposal 12494, Observation 2: NIRSPEC**

Thu Dec 18 20:00:32 GMT 2025

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | AT2025abao | RA: 00 38 48.6040 (9.7025167d)<br>Dec: +40 46 8.20 (40.76894d)<br>Equinox: J2000 | Epoch of Position: 2016 | |

*Comments:*
*Category=Star*
*Description=[Eruptive variables, K giants, M stars]*
*Extended=NO*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | SUB32 | F140X | NRSRAPIDD6 | 3 | 1 | 1 | 0.26 | 281633 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S200A1 | SUBS200A1 |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 3 | NONE |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G140H/F070LP | S200A1 | NRSRAPID | 42 | 3 | 1 | NONE | 3 | 9 | 603.13 | 281633 |
| 2 | G140H/F100LP | S200A1 | NRSRAPID | 42 | 3 | 2 | NONE | 3 | 9 | 603.13 | 281633 |
| 3 | G235H/F170LP | S200A1 | NRSRAPID | 42 | 3 | 3 | NONE | 3 | 9 | 603.13 | 281633 |
| 4 | G395H/F290LP | S200A1 | NRSRAPID | 42 | 3 | 4 | NONE | 3 | 9 | 603.13 | 281633 |

| Special Requirements | |
|---|---|
| | Between Dates 22-NOV-2025:00:00:00 and 23-JAN-2026:00:00:00 |

## Observation

**Proposal 12494, Observation 4: NIRSPEC_epoch2**

Thu Dec 18 20:00:32 GMT 2025

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | AT2025abao | RA: 00 38 48.6040 (9.7025167d)<br>Dec: +40 46 8.20 (40.76894d)<br>Equinox: J2000 | Epoch of Position: 2016 | |

*Comments:*
*Category=Star*
*Description=[Eruptive variables, K giants, M stars]*
*Extended=NO*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | SUB32 | F140X | NRSRAPIDD6 | 3 | 1 | 1 | 0.26 | 281633 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S200A1 | SUBS200A1 |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 3 | NONE |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | G140H/F070LP | S200A1 | NRSRAPID | 42 | 3 | 1 | NONE | 3 | 9 | 603.13 | 281633 |
| 2 | G140H/F100LP | S200A1 | NRSRAPID | 42 | 3 | 2 | NONE | 3 | 9 | 603.13 | 281633 |
| 3 | G235H/F170LP | S200A1 | NRSRAPID | 42 | 3 | 3 | NONE | 3 | 9 | 603.13 | 281633 |
| 4 | G395H/F290LP | S200A1 | NRSRAPID | 42 | 3 | 4 | NONE | 3 | 9 | 603.13 | 281633 |

| Special Requirements | Between Dates 14-JUL-2026:00:00:00 and 16-SEP-2026:00:00:00 |
|---|---|